PHYS 365 & PHYS 369
Midterm Exam

VERSION 01

October 21, 2026, 7:00-8:30 PM Time: 90 minutes

This is a closed-book exam containing a total of 20 equally weighted questions.

ONLY THE FOLLOWING ITEMS ARE ALLOWED ON YOUR DESK DURING THE EXAM:

1. THIS MULTIPLE-CHOICE QUESTION BOOKLET, which includes the exam questions, the formula sheet, and 3 blank pages. This booklet WILL **NOT** BE HANDED IN.

2. A BUBBLE SHEET used to answer multiple-choice questions. This will be handed in and it WILL BE GRADED.

3. A CALCULATOR. Only a Schulich-approved scientific calculator may be used during the exam.

4. A PEN OR PENCIL, used to black out your answers on the bubble sheet.

5. YOUR STUDENT ID CARD, used to check your identity during exam sign-in.

Students remaining after **75 minutes** must remain seated for the duration of the exam until permitted to leave by the exam supervisor.

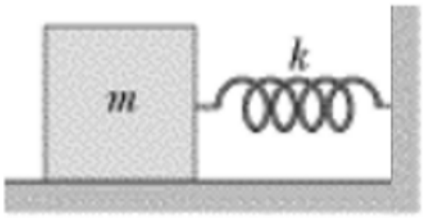
1. A mass is undergoing simple harmonic motion (SHM). At a particular instant, you know only its position x relative to equilibrium. Can you determine its velocity v from this information?

   A) Yes, because the velocity is uniquely determined by the position in SHM.
   B) Yes, because the velocity is always proportional to the displacement.
   C) No, because the mass could be moving in either direction at the same position.
   D) No, because velocity and displacement are completely unrelated in SHM.

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2. A spring with spring constant 160 N/m is connected to a block of mass 36.9 kg that is set oscillating over a frictionless floor. What is the frequency of the oscillations?

   A) 0.663 Hz
   B) 0.331 Hz
   C) 2.082 Hz
   D) 0.166 Hz



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3. In the Jubilee Auditorium Calgary there are speakers placed in the back close to the ceiling to help the back rows hear the show. On average the second balcony seats are $D_f = 42$ m away from the stage and $D_b = 8$ m away from the ceiling speakers. The speed of sound in air is $v = 340$ m/s. What time delay should I set between the stage microphones and back ceiling speakers, so the sound arrives at the audience at the same time?

   A) $\Delta t = 100$ ms
   B) $\Delta t = 23$ ms
   C) $\Delta t = 146$ ms
   D) $\Delta t = 122$ ms

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4. A simple harmonic oscillator consists of a block of mass 4.60 kg attached to a spring of spring constant 360 N/m. At $t = 0.61$ s, the position of the block is $x = 0.103$ m and the velocity of the block is $v = 3.900$ m/s. What is the amplitude of the oscillations?

   A) 0.325 m
   B) 0.621 m
   C) 0.453 m
   D) 0.103 m

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5. Select the TRUE statement about the energy of an object experiencing simple harmonic motion with frequency $f = 5$ Hz.

   A) The total energy E oscillates with frequency $f = 5$ Hz.
   B) The total energy E is a constant, but the potential energy U oscillates between zero and E 2.5 times per second.
   C) The total energy E is a constant, but the kinetic energy K oscillates between zero and E 5 times per second.
   D) The total energy E is a constant, but the potential energy U oscillates between zero and E 10 times per second.

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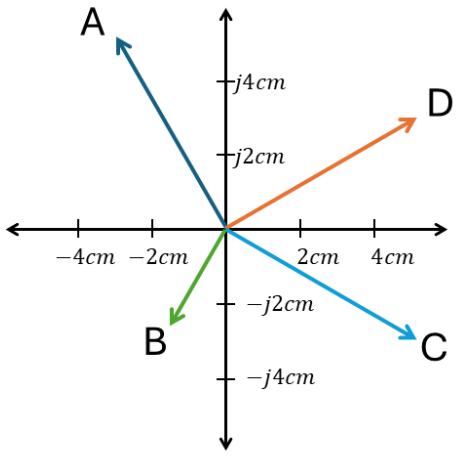
6. Two sinusoidal waves of the same wavelength travel in the same direction along a stretched string. For wave 1, $y_1 = 2.2$ mm and $\varphi_1 = 0°$. For wave 2, $y_2 = 4.4$ mm and $\varphi_2 = 72°$. What are the amplitude and phase constant of the resultant wave?

   A) 4.4 mm, 36.0°
   B) 5.49 mm, 49.6°
   C) 6.60 mm, 72.0°
   D) 5.49 mm, 36.0°

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7. Which of the phasors on the diagram represents a traveling wave described by the following equation?

$$y(x,t) = (6\text{ cm})\cos\left(2764t - 8.06x + \frac{13\pi}{6}\right)$$



   A) A
   B) B
   C) C
   D) D

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8. A 5.0 m long vibrating string produces a sound that is 30 dB at a location of 2.5 cm away from the string. What is the power per unit length P/ℓ emitted by the string?

A) 1.57×10^{-10} W/m
B) 7.85×10^{-10} W/m
C) 3.93×10^{-9} W/m
D) 7.85×10^{-9} W/m

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9. Three waves are represented by the following wave functions:

$$z_1(x,t) = 5\sin(4x - 4t)$$

$$z_2(x,t) = 3\sin(3x - 9t)$$

$$z_3(x,t) = 2\sin(3x - 4t)$$

What is the correct ranking of these waves according to their wave speeds?

A) $v_3 > v_1 > v_2$
B) $v_1 > v_2 = v_3$
C) $v_2 > v_3 > v_1$
D) $v_2 > v_1 = v_3$

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10. Nitrogen and oxygen gases have mean molecular masses of $M_{N_2} = 28.014$ g/mol and $M_{O_2} = 31.998$ g/mol, respectively. Air is a mixture of 78.08% nitrogen and 21.92% oxygen. At room temperature $T = 20°\text{C}$ what is the **difference** between the speed of sound in pure nitrogen versus air?

A) 343.68 m/s
B) 5.32 m/s
C) 10.71 m/s
D) 11.78 m/s

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11. Two cylindrical beams are made of the same material but one has twice the radius of the other $r_1 = 2r_2$. A percussive wave pulse (a hit with a hammer) on one end of the first beam travels to the other end in time t. Same percussive wave pulse travels through the second beam in twice the time. What is the relationship between the length of the beams?

A) $L_2 = 4L_1$

B) $L_2 = \sqrt{2}L_1$

C) $L_2 = L_1$

D) $L_2 = 2L_1$

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12. Consider a scenario when you send a traveling wave along a particular string by oscillating one end. If you then increase the frequency of oscillations while maintaining the same tension, which of the following statements is TRUE about the speed of the wave and its wavelength?

A) Speed increases, wavelength increases.
B) Speed is constant, wavelength increases.
C) Speed is constant, wavelength decreases
D) Speed increases, wavelength decreases.

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13. Three air columns, C_1 with both ends open, C_2 with one end open and one closed, and C_3 with both ends closed, all have the same length L. C_1 is excited to its 3rd harmonic, C_2 is excited to its 5th harmonic, and C_3 is excited to its 4th harmonic. What is the correct ranking of the 3 frequencies?

A) $C_2 > C_3 > C_1$
B) $C_1 > C_2 > C_3$
C) $C_3 > C_2 = C_1$
D) $C_3 > C_1 > C_2$

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14. Which of the following statements is TRUE about the wave number of a travelling harmonic wave on a string?

A) The wave number is inversely related to its wavelength.
B) The wave number is related to the number of waves on the string.
C) The wave number depends on the amplitude of the wave.
D) The wave number is independent to the wave's wavelength.

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15. At a certain harbor, the tides cause the ocean surface to rise and fall a distance d (from highest level to lowest level) in simple harmonic motion, with a period of 12.4 h. How long does it take for the water to fall a distance $0.25d$ from its highest level?

A) 1.55 h
B) 2.07 h
C) 2.48 h
D) 3.1 h

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16. Two people are in a long narrow corridor: one hits a tuning fork and the other hears it at a volume of normal conversation. The speed of sound in air is 343.69 m/s and the frequency of the tuning fork is 440 Hz. The pressure amplitude of normal conversation is $\Delta p_{max} = 0.02$ Pa. If the density of the air in the corridor is $\rho = 1.2\ \text{kg/m}^3$, what is the maximum speed at which the air molecules move?

A) 4.85×10^{-5} m/s
B) 1.75×10^{-8} m/s
C) 7.23×10^{-6} m/s
D) 1.10×10^{-7} m/s

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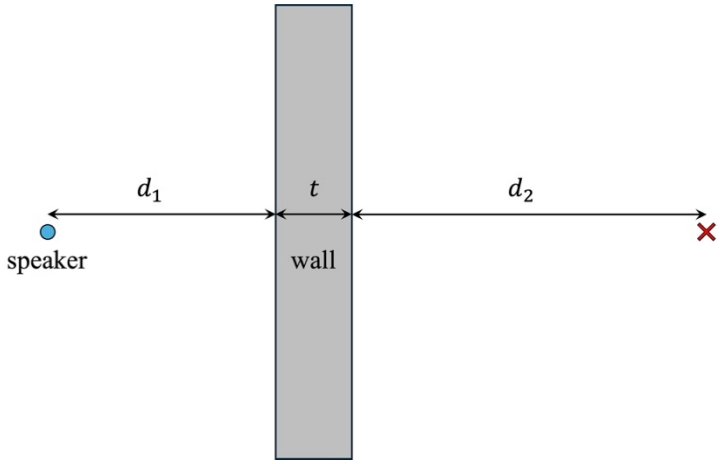
17. The low E string on a guitar should be tuned to E_2 at 82 Hz but you find that it has been tuned one octave lower to E_1 at 41 Hz. By what factor must you increase the tension in the string to tune it properly?

A) 2
B) 16
C) $\sqrt{2}$
D) 4

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18. A 40 W speaker acting as a point source of sound is $d_1 = 3.0$ m away from a concrete wall that is $t = 1.0$ m thick and has an acoustic attenuation coefficient of $\alpha = 0.13\ \text{cm}^{-1}$, as shown in the diagram. How far away from the wall on the other side, d_2, is the sound level of the speaker 50 dB?

A) 1.8 m
B) 3.8 m
C) 4.5 m
D) 8.5 m



The following scenario applies to Questions 19 and 20

An ultrasound device emits acoustic waves at 6.50 MHz from a probe placed on the skin, which has an acoustic impedance of 1.68×10^6 Pa · s/m. A gel with acoustic impedance 1.50×10^6 Pa · s/m is used between the skin and the probe so that most of the wave is transmitted through the skin, instead of reflecting off due to an air gap, which has a very low impedance of 415 Pa · s/m.

19. What percentage of the acoustic wave's energy would be **reflected off** the skin during an ultrasound if **no gel** were used?

A) 99.90%
B) 99.79%
C) 94.50%
D) 89.99%

20. What percentage of the acoustic wave's energy is **transmitted through** the skin during an ultrasound when the gel is used?

A) 94.34%
B) 99.68%
C) 89.00%
D) 105.7%

PHYS 365 & 369 Midterm Exam Formula Sheet

OSCILLATIONS

$x(t) = x_m \cos(\omega t + \phi)$	$\omega = 2\pi f = \frac{2\pi}{T}$	$\omega = \sqrt{\frac{k}{m}}$	$\omega = \sqrt{\frac{g}{\ell}}$	$T = 2\pi\sqrt{\frac{\ell}{g}}$

WAVES

$\frac{\partial^2 y}{\partial^2 t} = v^2 \frac{\partial^2 y}{\partial x^2}$	$y(x,t) = f(x \pm vt)$	$y(x,t) = y_m \cos(kx \pm \omega t + \phi)$	
$v = \frac{\omega}{k} = f\lambda$	$k = \frac{2\pi}{\lambda}$	$v = \frac{c}{n}$	$\lambda = \frac{\lambda_0}{n}$
$v = \sqrt{\frac{\tau}{\mu}}$	$v = \sqrt{\frac{\gamma RT}{M}}$	$v = \sqrt{\frac{Y}{\rho}}$	$v = \sqrt{\frac{B}{\rho}}$
$Z = \sqrt{\mu\,\tau} = \frac{\tau}{v} = \mu v$	$Z = \sqrt{\rho B} = \rho v$	$P_{avg} = \frac{1}{2}\mu v \omega^2 y_m^2$	$\Delta p_m = (v\rho\omega)s_m$
$\Delta p = \Delta p_m \sin(kx - \omega t)$	$I = \frac{1}{2}\rho v \omega^2 s_m^2$	$I = \frac{P}{4\pi R^2}$	$I = \frac{P}{A}$
$\beta = (10\ dB)\log_{10}\left(\frac{I}{I_0}\right)$	$\Delta\beta = (10\ dB)\log_{10}\left(\frac{I_2}{I_1}\right)$	$I(r_2) = \left(\frac{r_1}{r_2}\right)^{n-1} I(r_1)e^{-\alpha\Delta r}$	
$R = \frac{Z_1 - Z_2}{Z_1 + Z_2}$	$T = \frac{2Z_1}{Z_1 + Z_2}$	$B = -\frac{\Delta p}{\left(\frac{\Delta V}{V}\right)}$	$Y = \frac{\left(\frac{F}{A}\right)}{\left(\frac{\Delta L}{L}\right)}$
$R_e = R^2$	$T_e = \frac{Z_2}{Z_1}T^2$	$\Delta p = -B\frac{\partial s}{\partial x}$	

WAVE INTERFERENCE

$y'(x,t) = \left[2y_m \cos\left(\frac{1}{2}\phi\right)\right]\sin\left(kx - \omega t + \frac{1}{2}\phi\right)$		$y'(x,t) = [2y_m \sin(kx)]\cos(\omega t)$	
$\Delta\phi = \frac{\Delta L}{\lambda}2\pi$	$f_{beat} = f_1 - f_2$	$f' = f\frac{v \pm v_D}{v \mp v_S}$	$f' = f\sqrt{\frac{c \pm v}{c \mp v}}$
Constructive interference $\frac{\Delta L}{\lambda} = 0,\ 1,\ 2, \ldots$		Destructive interference $\frac{\Delta L}{\lambda} = 0.5,\ 1.5,\ 2.5, \ldots$	
Taut string or pipe open at both ends $f_n = \frac{v}{\lambda_n} = \frac{nv}{2L}$ for $n = 1,\ 2,\ 3, \ldots$		Pipe one end closed, one end open $f_n = \frac{v}{\lambda_n} = \frac{nv}{4L}$ for $n = 1,\ 3,\ 5, \ldots$	
$d\sin(\theta) = m\lambda$ for $m = 0, 1, 2, \ldots$ (maxima-bright fringes)			
$d\sin(\theta) = \left(m + \frac{1}{2}\right)\lambda$ for $m = 0, 1, 2, \ldots$ (minima-dark fringes)			

$2t = m\frac{\lambda_0}{n}, \quad m = 1, 2, 3 \ldots$		$2t = \left(m - \frac{1}{2}\right)\frac{\lambda_0}{n}, \quad m = 1, 2, 3 \ldots$	
Reflection	Reflection phase shift	$a\sin(\theta) = m\lambda$ for $m = 1, 2, 3, \ldots$ (minima)	
Off lower index	0		
Off higher index	0.5 wavelength		
$I = 4I_0\cos^2\left(\frac{1}{2}\phi\right)$	$\phi = \frac{2\pi d}{\lambda}\sin(\theta)$	$\theta_R = \sin^{-1}\left(\frac{1.22\lambda}{d}\right)$	
CONSTANTS			
$c = 3.00 \times 10^8\ \frac{\text{m}}{\text{s}}$	$R = 8.314\ \frac{\text{J}}{\text{mol}\cdot\text{K}}$	$M_{air} = 28.8\ \frac{\text{g}}{\text{mol}}$	$g = 9.81\ \frac{\text{m}}{\text{s}^2}$
$v_{sound\ at\ 20℃} = 343\ \frac{\text{m}}{\text{s}}$	$T_K = T_{℃} + 273.15$	$\gamma_{monoatomic} = \frac{5}{3}$	
$I_0 = 1 \times 10^{-12}\ \frac{\text{W}}{\text{m}^2}$	$1\ \text{atm} = 1.01 \times 10^5\ \text{Pa}$	$\gamma_{diaatomic} = \frac{7}{5}$	
MATHEMATICAL FORMULAE			
$\cos(\alpha) \pm \cos(\beta) = 2\cos\left(\frac{1}{2}(\alpha \pm \beta)\right)\cos\left(\frac{1}{2}(\alpha \mp \beta)\right)$			
$\sin(\alpha) \pm \sin(\beta) = 2\sin\left(\frac{\alpha \pm \beta}{2}\right)\cos\left(\frac{\alpha \mp \beta}{2}\right)$			
$\cos(\alpha \pm \beta) = \cos(\alpha)\cos(\beta) \mp \sin(\alpha)\sin(\beta)$		$\tilde{A} = Ae^{j\theta} = A(\cos(\theta) + j\sin(\theta))$	
$\sin(\alpha \pm \beta) = \sin(\alpha)\cos(\beta) \pm \cos(\alpha)\sin(\beta)$		$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	
$\sin(-\alpha) = -\sin(\alpha)$		$\cos\left(\alpha \pm \frac{\pi}{2}\right) = \mp\sin(\alpha)$	
$\cos(-\alpha) = \cos(\alpha)$		$\sin\left(\alpha \pm \frac{\pi}{2}\right) = \pm\cos(\alpha)$	